

DAY 8

Jupyter day8 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Anaconda Toolbox

EDA

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[2]: df=sns.load_dataset("tips")

[30]: print(df.head(20))
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
5	25.29	4.71	Male	No	Sun	Dinner	4
6	8.77	2.00	Male	No	Sun	Dinner	2
7	26.88	3.12	Male	No	Sun	Dinner	4
8	15.04	1.96	Male	No	Sun	Dinner	2
9	14.78	3.23	Male	No	Sun	Dinner	2
10	10.27	1.71	Male	No	Sun	Dinner	2
11	35.26	5.00	Female	No	Sun	Dinner	4
12	15.42	1.57	Male	No	Sun	Dinner	2
13	18.43	3.00	Male	No	Sun	Dinner	4
14	14.83	3.02	Female	No	Sun	Dinner	2
15	21.58	3.92	Male	No	Sun	Dinner	2
16	10.33	1.67	Female	No	Sun	Dinner	3
17	16.29	3.71	Male	No	Sun	Dinner	3
18	16.97	3.50	Female	No	Sun	Dinner	3
19	20.65	3.35	Male	No	Sat	Dinner	3

```
[4]: print(df.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        -
0   total_bill    244 non-null    float64
1   tip           244 non-null    float64
..
```

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```
[5]: print(df.columns)

Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')

[6]: print(df.describe())
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
*[10]: #Descriptive Statistics
print(df['tip'].describe()) #for a particular column

count    244.000000
mean      2.998279
std       1.383638
min       1.000000
25%       2.000000
50%       2.900000
75%       3.562500
max       10.000000
Name: tip, dtype: float64

[14]: #HISTOGRAM

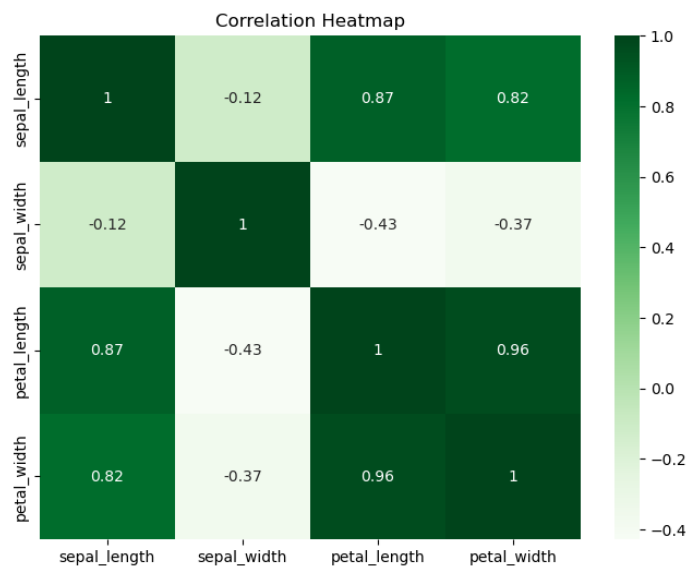
sns.histplot(df['total_bill'],bins=20,kde=True)
plt.title("Distribution of Total bill")
plt.show()
```

```
[79]: from statsmodels.stats.outliers_influence import variance_inflation_factor
x=df.drop("species",axis=1)
vif=pd.DataFrame()
vif["Feature"]=x.columns
vif["VIF"]=[
    variance_inflation_factor(x.values,i)
    for i in range(x.shape[1])
]
print(vif)
```

	Feature	VIF
0	sepal_length	262.969348
1	sepal_width	96.353292
2	petal_length	172.960962
3	petal_width	55.502060

```
•[83]: #REMOVE A HIGH-VIF FEATURE
#suppose we remove petal_length
x_new=x.drop("sepal_length",axis=1)
vif_after=pd.DataFrame()
vif_after["Feature"]=x_new.columns
vif_after["VIF"]=[
    variance_inflation_factor(x_new.values,i)
    for i in range(x_new.shape[1])
]
print(vif_after)
```

	Feature	VIF
0	sepal_width	5.856965
1	petal_length	62.071308
2	petal_width	43.292574



```
[79]: from statsmodels.stats.outliers_influence import variance_inflation_factor
x=df.drop("species",axis=1)
vif=pd.DataFrame()
vif["Feature"]=x.columns
vif["VIF"]=[
    variance_inflation_factor(x.values,i)
    for i in range(x.shape[1])
]
print(vif)
```